

Beyond the Training Distribution

Taxonomy of Reasoning Generalization in Large Language Models

Reasoning Generalization in LLMs

1. Training for Generalization

Pre-training

Training exposure and capability boundaries

controlled pre-training, mid-training, and RL exposure • Interplay of Training Stages (2025)

Mid-training

Continued pre-training and curricula

verified state traces and positional curricula • TAIL (2025); Randomized YaRN; Left-Shift (2026)

Post-training

Supervised adaptation

SFT and distillation: expose transferable procedures

composable traces, rule targets, negative evidence • Composable CoT (2025); Theorem-SFT; IGA (2026)

RL & reward design

Select and compose procedures with feedback

process rewards and skill composition • GraphPRM; LogicPuzzleRL; f $\square$ g RL (2025); GRAIN (2026)

Hybrid & self-improvement

Combine learning stages and generated experience

iterative self-training and stage handoffs • Self-Improving Transformers (2025); STEPS; CALM (2026)

2. Inference for Generalization

Prompt & decompose

Elicit a reusable procedure

easy-to-hard decomposition, constraint transfer, routed hints • Least-to-Most; CFR; Test-Time Hinting

Sample & search

Explore and aggregate candidate paths

candidate coverage, error-localized branching, valid-prefix reuse • TTEL cross-domain transfer (2026)

Verify, repair & stop

Transfer validity detection and inference control

unseen benchmarks, cross-domain monitors, prefix reuse • Validity Detector; TTEL; UPAIR (2026)

Tools & memory

Externalize exact operations and reusable experience

tool-based length transfer, cross-task memory, modular online reuse • To Infinity; ReasoningBank; MILES

3. Architecture for Generalization

Recurrent depth

Shared transition, variable compute, learned halting

looped transformers, depth recurrence, stochastic stopping • Loop-Think-Generalize; Equilibrium Reasoners (2026)

Position & locality

Relative coordinates and task-aligned neighborhoods

task-aligned coordinates and local updates • Position Coupling; Structural Symmetry (2024); RFS (2026)

Modules & symbols

Reusable subroutines and exact state

structured scratchpads, causal graphs, symbolic state • Arithmetic Transformers (2024); AGEL-Comp (2026)

Latent & equilibrium

Iterative hidden-state computation

recursive latent reasoning, energy minimization, attractors, stable dynamics • Think Shallow, Solve Deep (2026)

4. Analysis of Generalization

4.1 Behavioral observations

Shift patterns

surface, composition, length/depth, difficulty, domain, language, modality

GSM-Plus; GSM-Symbolic; CGGC (2024) • General365; XDomainBench (2026)

Scaling patterns

thresholds, grokking, easy-to-hard transfer, overthinking, compute-cost curves

MathGAP (2024); Depth Generalization (2025) • test whether added compute expands capability

Validity controls

contamination, sealed generators, fresh symbols, counterfactuals, matched compute

modeling (2024); sealed generators; fresh symbols • separate recall from procedure transfer

4.2 Mechanisms & explanations

4.2.1 Empirical mechanisms

circuits, representation alignment, routing, learning dynamics, causal intervention

CoT Composition; Pattern Matching Limits (2025) • Shared Circuits; Self-Patching (2026)

4.2.2 Theoretical analysis

expressivity vs. learnability; RASP/languages; equivariance; bounds; impossibility

Provable Length & Composition; How Far (2024) • CoT Guarantees; Algebraic Decomposition